

LAB ACTIVITY: Hardening the Gateway & WAN Integration

1. Hole Punching: Explain how the Gateway was able to connect to the Client without you touching the router settings.

- The Gateway established a direct connection to the Client without manual router configuration by utilizing Tailscale's automated STUN (Session Traversal Utilities for NAT) and hole-punching techniques. Because both the Gateway and the Client initially establish outbound connections to a central Tailscale coordination server, they can exchange public IP addresses and dynamically mapped UDP ports. By sending synchronized UDP packets to each other simultaneously, the respective routers are tricked into opening dynamic pathways, allowing direct peer-to-peer traffic to pass through the NAT firewalls without requiring manual port forwarding rules..

2. Traffic Analysis: Why is it more secure to use an Overlay Network than to open Port 22 on a public-facing router?

- Using an overlay network is significantly more secure because it completely hides the server from the public internet, eliminating exposure to automated credential-stuffing bots and vulnerability scanners. When you open standard Port 22 on a public router, anyone on the internet can see the open port and attempt to exploit it. In contrast, an overlay network encrypts all traffic end-to-end and requires cryptographic node authentication; only pre-authorized devices belonging to your private network can even attempt to connect to the service, rendering it completely invisible to outsiders.

3. The Hotspot Factor: How did the latency change when you switched to a mobile hotspot or alternative network?

- When switching from a standard local network connection to a mobile hotspot, network latency generally increases, resulting in a higher round-trip time (RTT) for data packets. This happens because cellular data networks involve wireless transmission over longer distances to cell towers, which introduces overhead from signal modulation, environmental interference, and cellular carrier routing. While local Wi-Fi networks typically offer low, single-digit millisecond latency, a mobile hotspot introduces a noticeable delay—often jumping to 40ms or higher—which can make live SSH terminal sessions feel slightly less responsive.